Right triangles and trigonometry answer key

#	ID	Answer	Difficulty
1	359c6a4b-24df-4f06-8a59-45dbbbb00833	C	Easy
2	21685f5b-1a79-4984-81ee-932431a02a7a	B	Easy
3	2a386242-ddf5-4a21-9745-32467fe7a279	C	Easy
4	bc3176be-4d2c-4e9b-89aa-312963604d41	C	Easy
5	924b44e4-dcc1-47ff-ad3e-564a5a1991ef	A	Easy
6	5466f88b-12d1-4e2d-a8ae-304ba95ffcae	D	Easy
7	aefa5414-12c1-4240-9455-47ff78e9896d	C	Easy
8	a9d7a8cf-fa0b-4706-ba20-8d948b2ba69d	C	Easy
9	8d9a8a64-b22e-4435-a19b-1ac97f998d7a	B	Medium
10	aa7ee350-2124-4fd5-afe1-f10a1f785d05	B	Medium
11	b479bf47-357b-4283-81e0-02fa0f919256	B	Medium
12	09170-DC	?	Medium
13	19da9e4a-267f-405a-bab5-768d4d5c4c29	B	Medium
14	8d60c337-175f-44ef-817b-67041302a3bd	B	Medium
15	c15bdfb8-01a8-4d34-9cc6-822ca3108767	B	Medium
16	1673e5c1-c49a-4299-ab8c-5435403ce71d	D	Medium
17	08251-DC	C	Medium
18	076bf06d-2022-480c-ada2-6ea25ff29991	C	Medium
19	2ea4a48a-1af7-4130-bdcd-0c54f14d34be	45	Medium
20	7e480092-1ead-484d-8b6f-5c6417fa9d9f	A	Medium
21	6a004696-92f8-48b5-9534-0de1a1176958	C	Medium
22	ef234bbf-719d-4bc2-a841-937a532d135b	B	Medium

23	c9527c06-e018-47ad-b3bf-4779309e09b7	45	Medium
24	62bc2dfe-0a0c-45ec-a764-c5f84ddbce67	B	Medium
25	04175-DC	B	Medium
26	137892dd-a025-491a-a463-42740b162c96	A	Medium
27	c0b57dc4-8482-4d44-a972-dc8f51d858a9	565	Medium
28	49178e05-53a1-4bcb-954a-708fc5b770ff	D	Hard
29	9312f237-66e1-4380-a28d-185ceec5d2ab	.3928, .3929, 11/28	Hard
30	ddfe8011-1fec-4e7d-81bb-e842017e4525	.14, 7/50	Hard
31	35758801-d927-43ce-8662-b8d768411b28	38	Hard
32	07822-DC	C	Hard
33	e8c34144-2ff1-46f4-99ab-36c137fcb022	36	Hard
34	8c6f39d2-0804-4eb5-8c97-b801dcd8a2e8	D	Hard
35	d8a83c5c-3c86-4caa-91c0-283aa8c5d6b0	C	Hard
36	575135d8-2a82-4346-993f-f5d9b5921696	12	Hard
37	889dae43-01c0-449b-abd7-1a385a85c6a7	A	Hard
38	5bf76a41-5cb0-4d02-8d3a-4c2fc8668b3a	C	Hard
39	09173-DC	?	Hard
40	a3d7bd43-4b60-4b1e-a039-96f3769a34f2	A	Hard
41	4363b40f-7867-454b-a423-9aa19a6be480	66	Hard
42	5582e974-2bdf-4b46-a406-07c130db3b37	C	Hard
43	383c05a2-4e2a-4c4e-8121-840ca478825a	D	Hard
44	081f3aa6-88ed-4da7-bc3a-e7bbfa7b4fca	284/3, 94.66, 94.67	Hard
45	00615-DC	?	Hard
46	af8e7cb1-d233-45ae-8811-379283e1e3a4	C	Hard
47	e8f7d348-9106-4f42-a41c-8400c1758a07	338	Hard
48	028928-DC	D	Hard

49	60ba6bab-7d16-46f1-9882-e80e2dd3577c	D	Hard
50	7dae086f-c99c-48f6-b4ad-68e5e1008035	2.4, 12/5	Hard
51	ccc97dea-b103-4868-86e4-80b756cd63d7	B	Hard
52	16428dd4-6b32-4580-ace2-d2dbf0505a76	B	Hard
53	8ee41a37-b82e-4974-ab8e-ab0531faca66	D	Hard
54	c66e94ab-e172-4e0b-9609-f3b62ef3f90d	.6956, .6957, 16/23	Hard
55	05777-DC	?	Hard
56	13cc44bf-f2ac-4f02-ac05-440e4b894f96	.8823, .8824, 15/17	Hard
57	1dd30616-5d7f-4401-8dd7-811d477d90d8	104	Hard
58	2124991b-31a9-4ce3-a963-094142d78f70	B	Hard
59	3e66c309-612d-4f4e-90d5-3132cd7909e6	.9811, 52/53	Hard
60	fa02652e-89a4-4784-990e-fd5f06d65862	A	Hard
61	029664-DC	?	Hard
62	988fed36-4761-4d7b-962b-35933cf1b89e	6.5, 13/2	Hard
63	2ec14ac8-bf5b-46cc-8993-b907b006b31c	.7241, 21/29	Hard
64	84b26634-2244-45e6-8504-ea367a0f2049	B	Hard
65	05758-DC	?	Hard
66	1ef8e9cb-4e5c-441a-9395-222ae1a20a61	D	Hard
67	ae6da77a-4de7-4375-a532-b00350663a9f	113	Hard
68	5e8901b5-d7f3-45e7-8324-de2b20574706	C	Hard
69	4959ea4c-8faf-4a7c-9ef9-60dc6d45d751	C	Hard

Rationales

▼ 1 — C

Choice C is correct. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the side opposite the angle to the length of the shorter side adjacent to the

angle. In the triangle shown, the length of the side opposite the angle with measure $x°$ is 26 units and the length of the side adjacent to the angle with measure $x°$ is 7 units. Therefore, the value of $\tan x°$ is $\frac{26}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2 — B

Choice B is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length t and legs with lengths r and s, then $r^2 + s^2 = t^2$. It's given in the right triangle shown that the legs have lengths of 9 and 6 and the hypotenuse has a length of c. Substituting 9 for r, 6 for s, and c for t in $r^2 + s^2 = t^2$ yields $9^2 + 6^2 = c^2$, or $117 = c^2$. Taking the square root of both sides of this equation yields $\pm\sqrt{117} = c$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{117}$, which is approximately equal to 10.8167. Of the choices, 10.8 is the closest to the value of c.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 3 — C

Choice C is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. In the right triangle shown, it's given that the length of the side opposite the angle with measure $x°$ is 63 units and the length of the hypotenuse is 73 units. Therefore, the value of $\sin x°$ is $\frac{63}{73}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 4 — C

Choice C is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse. For the given right triangle, the lengths of the legs are 4 and b, and the length of the hypotenuse is 19. Substituting 4 for a and 19 for c in the equation $a^2 + b^2 = c^2$ yields $4^2 + b^2 = 19^2$. Thus, the relationship between the side lengths of the given triangle is $4^2 + b^2 = 19^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 5 — A

Choice A is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to that angle to the length of the hypotenuse. It's given that angle C is a right angle in triangle ABC and that angle F is a right angle in triangle DEF. Therefore, $\cos B$ is equal to the ratio of the length of side BC to the length of side AB, and $\cos E$ is equal to the ratio of the length of side EF to the length of side DE. It's also given that triangle ABC is similar to triangle DEF, where angle A corresponds to angle D. Since similar triangles have proportional side lengths, $\frac{BC}{AB} = \frac{EF}{DE}$. Therefore, the value of $\cos B$ is equal to the value of $\cos E$. Since $\cos B = \frac{1}{22}$, the value of $\cos E$ is $\frac{1}{22}$.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 6 — D

Choice D is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $c^2 = a^2 + b^2$. In the right triangle shown, the hypotenuse has length c and the legs have lengths a and b. It's given that $a = 4$ and $b = 5$. Substituting 4 for a and 5 for b in the Pythagorean theorem yields $c^2 = 4^2 + 5^2$. Taking the square root of both sides of this equation yields $c = \pm\sqrt{4^2 + 5^2}$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{4^2 + 5^2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 7 — C

Choice C is correct. The Pythagorean theorem states that for a right triangle, $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs and c represents the length of the hypotenuse. It's given that the right triangle shown has legs with lengths 12 and 3 and a hypotenuse with length c. Substituting 12 for a and 3 for b in the formula $a^2 + b^2 = c^2$ yields $12^2 + 3^2 = c^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 8 — C

Choice C is correct. The Pythagorean theorem states that for a right triangle, $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs of the triangle and c represents the length of its hypotenuse. In the triangle shown, the legs have lengths of 3 and 7. Substituting 3 for a and 7 for b in the equation $a^2 + b^2 = c^2$ yields $3^2 + 7^2 = c^2$, which is equivalent to $9 + 49 = c^2$, or $58 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{58} = c$. Thus, the value of c is approximately 7.6. Therefore, of the given choices, 7.6 is the closest to the length of the triangle's hypotenuse.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 9 — B

Choice B is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to angle A is 41, and the length of the hypotenuse is 42. Therefore, $\cos A = \frac{41}{42}$.

Choice A is incorrect. This is the value of $\frac{1}{\cos A}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 10 — B

Choice B is correct. It's given that in $\triangle ABC$, $\angle B$ is a right angle. Therefore, $\triangle ABC$ is a right triangle, and $\overline{AC}$ is the hypotenuse of the triangle. It's also given that $\cos A = \frac{3}{5}$. Since the cosine of an acute angle in a right triangle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse, the ratio of the length of $\overline{AB}$ to the length of $\overline{AC}$ is 3 to 5. It follows that the length of $\overline{AB}$ can be represented as $3a$ and the length of $\overline{AC}$ can be represented as $5a$, where a is a constant. The Pythagorean theorem states that the sum of the squares of the length of the legs of a right triangle is equal to the square of the length of its hypotenuse, so it follows that $AB^2 + BC^2 = AC^2$. Substituting $3a$ for AB and $5a$ for AC in this equation yields $3a^2 + BC^2 = 5a^2$, or $9a^2 + BC^2 = 25a^2$. Subtracting $9a^2$ from both sides of this equation yields $BC^2 = 16a^2$, or $BC = 4a$. It follows that the ratio of the length of $\overline{AB}$ to the length of $\overline{BC}$ is 3 to 4. Let x represent the length, in millimeters, of $\overline{AB}$. It's given that the length of $\overline{BC}$ is 136 millimeters. Since the ratio of the length of $\overline{AB}$ to the length of $\overline{BC}$ is 3 to 4, $\frac{x}{136} = \frac{3}{4}$. Multiplying both sides of this equation by 136 yields $x = \frac{3(136)}{4}$, or $x = 102$. Therefore, the length of $\overline{AB}$ is 102 millimeters.

Choice A is incorrect. This is the scale factor by which the 3 to 4 to 5 ratio is multiplied that results in the side lengths of $\triangle ABC$.

Choice C is incorrect. This is the length of $\overline{BC}$, not the length of $\overline{AB}$.

Choice D is incorrect. This is the length of $\overline{AC}$, not the length of $\overline{AB}$.

▼ 11 — B

Choice B is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. It's given that the right triangle has legs with lengths of 28 centimeters and 20 centimeters. Let c represent the length of this triangle's hypotenuse, in centimeters. Therefore, by the Pythagorean theorem, $28^2 + 20^2 = c^2$, or $1{,}184 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{1{,}184} = c$, or $4\sqrt{74} = c$. Therefore, the length of this triangle's hypotenuse, in centimeters, is $4\sqrt{74}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the square of the length of the triangle's hypotenuse.

▼ 12 — ?

The correct answer is 4. Triangle PQR has given angle measures of 30° and 90°, so the third angle must be 60° because the measures of the angles of a triangle sum to 180°. For any special right triangle with angles measuring 30°, 60°, and 90°, the length of the hypotenuse (the side opposite the right angle) is 2x, where x is the length of the side opposite the 30° angle. Segment PQ is opposite the 30° angle. Therefore, 2(PQ) = 8 and PQ = 4.

▼ 13 — B

Choice B is correct. The sine of any acute angle is equal to the cosine of its complement. It's given that in right triangle *RST*, the sum of the measures of angle *R* and angle *S* is 90 degrees. Therefore, angle *R* and angle *S* are complementary, and the value of $\sin R$ is equal to the value of $\cos S$. It's given that the value of $\sin R$ is $\frac{\sqrt{15}}{4}$, so the value of $\cos S$ is also $\frac{\sqrt{15}}{4}$.

Choice A is incorrect. This is the value of $\tan S$.

Choice C is incorrect. This is the value of $\frac{1}{\cos S}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan S}$.

▼ 14 — B

Choice B is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. The hypotenuse of a right triangle is the side opposite the right angle. In right triangle *ABC*, side *BC* is the side opposite angle *A* and side *AB* is the hypotenuse. It's given that the length of side *BC* is 35 units and the length of side *AB* is 171 units. Therefore, the value of $\sin A$ is $\frac{35}{171}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the ratio of the length of the hypotenuse to the length of the side opposite angle *A* rather than the ratio of the length of the side opposite angle *A* to the length of the hypotenuse.

Choice D is incorrect. This is the length of the hypotenuse rather than $\sin A$.

▼ 15 — B

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that a right triangle has legs with lengths of 11 centimeters and 9 centimeters. Substituting 11 for a and 9 for b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 11^2 + 9^2$, which is equivalent to $c^2 = 121 + 81$, or $c^2 = 202$. Taking the square root of each side of this equation yields $c = \pm\sqrt{202}$. Since c represents a length, c must be positive. Therefore, the length of the triangle's hypotenuse, in centimeters, is $\sqrt{202}$.

Choice A is incorrect. This is the result of solving the equation $c^2 = 112 + 92$, not $c^2 = 11^2 + 9^2$.

Choice C is incorrect. This is the result of solving the equation $c2 = 112 + 92$, not $c^2 = 11^2 + 9^2$.

Choice D is incorrect. This is the result of solving the equation $c = 11^2 + 9^2$, not $c^2 = 11^2 + 9^2$.

▼ 16 — D

Choice D is correct. The sine of an angle is equal to the cosine of its complementary angle. In the triangle shown, angle Z is a right angle; thus, angles X and Y are complementary angles. Therefore, $\cos Y$ has the same value as $\sin X$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 17 — C

Choice C is correct. Angle A is an acute angle in a right triangle, so the value of tan(A) is equivalent to the ratio of the length of the side opposite angle A, 20, to the length of the nonhypotenuse side adjacent to angle A, 21. Therefore, $\tan(A) = \frac{20}{21}$.

Choice A is incorrect. This is the value of sin(A). Choice B is incorrect. This is the value of cos(A). Choice D is incorrect. This is the value of tan(B).

▼ 18 — C

Choice C is correct. The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. It's given that one leg of a right triangle has a length of 43.2 millimeters. It's also given that the hypotenuse of the triangle has a length of 196.8 millimeters. Let b represent the length of the other leg of the triangle, in millimeters. Therefore, by the Pythagorean theorem, $43.2^2 + b^2 = 196.8^2$, or $1{,}866.24 + b^2 = 38{,}730.24$. Subtracting $1{,}866.24$ from both sides of this equation yields $b^2 = 36{,}864$. Taking the positive square root of both sides of this equation yields $b = 192$. Therefore, the length of the other leg of the triangle, in millimeters, is 192.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 19 — 45

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 15. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are $45°$. It follows that the value of x is 45.

▼ 20 — A

Choice A is correct. Since the two acute angles have the same measure and the third angle is a right angle, the triangle shown is an isosceles right triangle. In an isosceles right triangle, the two legs have the same length. The figure shows that the length of one leg of the triangle is 24 and the length of the other leg of the triangle is x. It follows that the value of x is 24.

Choice B is incorrect. This is the measure, in degrees, of one of the angles shown.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 21 — C

Choice C is correct. It's given that the length of $\overline{DE}$ is 147, and the length of $\overline{AE}$ is 245. Since angle D is a right angle, triangle ADE is a right triangle with hypotenuse $\overline{AE}$ and legs $\overline{AD}$ and

$\bar{DE}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting AD for a, 147 for b, and 245 for c in the equation $a^2 + b^2 = c^2$ yields $AD^2 + 147^2 = 245^2$. Subtracting 147^2 from both sides of this equation yields $AD^2 = 245^2 - 147^2$. Evaluating this difference yields $AD^2 = 60{,}025 - 21{,}609$, or $AD^2 = 38{,}416$. Taking the positive square root of both sides of this equation yields $AD = 196$. It's given that $\bar{BC}$ is parallel to $\bar{DE}$, and $\bar{AE}$ intersects $\bar{BC}$ and $\bar{DE}$. It follows that angle ACB is congruent to angle AED. Since angle A is common to triangles ABC and ADE, it follows that triangle ABC is similar to triangle ADE. In triangle ABC, $\tan x°$ is the ratio of the leg opposite angle ACB to the leg adjacent to angle ACB, so $\tan x° = \frac{AB}{BC}$. Since triangle ABC is similar to triangle ADE and similar triangles have proportional side lengths, it follows that $\frac{AB}{BC} = \frac{AD}{DE}$. Therefore, $\tan x° = \frac{AD}{DE}$. Substituting 196 for AD and 147 for DE in this equation yields $\tan x° = \frac{196}{147}$. Therefore, the value of $\tan x°$ is $\frac{196}{147}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\cos x°$, not $\tan x°$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 22 — B

Choice B is correct. A rectangle's diagonal divides a rectangle into two congruent right triangles, where the diagonal is the hypotenuse of both triangles. It's given that the length of the diagonal is $5\sqrt{17}$ and the length of the rectangle's shorter side is 5. Therefore, each of the two right triangles formed by the rectangle's diagonal has a hypotenuse with length $5\sqrt{17}$, and a shorter leg with length 5. To calculate the length of the longer leg of each right triangle, the Pythagorean theorem, $a^2 + b^2 = c^2$, can be used, where a and b are the lengths of the legs and c is the length of the hypotenuse of the triangle. Substituting 5 for a and $5\sqrt{17}$ for c in the equation $a^2 + b^2 = c^2$ yields $5^2 + b^2 = 5\sqrt{17}^2$, which is equivalent to $25 + b^2 = 2517$, or $25 + b^2 = 425$. Subtracting 25 from each side of this equation yields $b^2 = 400$. Taking the positive square root of each side of this equation yields $b = 20$. Therefore, the length of the longer leg of each right triangle formed by the diagonal of the rectangle is 20. It follows that the length of the rectangle's longer side is 20.

Choice A is incorrect and may result from dividing the length of the rectangle's diagonal by the length of the rectangle's shorter side, rather than substituting these values into the Pythagorean theorem.

Choice C is incorrect and may result from using the length of the rectangle's diagonal as the length of a leg of the right triangle, rather than the length of the hypotenuse.

Choice D is incorrect. This is the square of the length of the rectangle's longer side.

▼ 23 — 45

The correct answer is 45. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of 32. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are 45°. It follows that the value of *x* is 45.

▼ 24 — B

Choice B is correct. If two triangles are similar, then their corresponding angles are congruent. It's given that right triangle *FGH* is similar to right triangle *JKL* and angle *F* corresponds to angle *J*. It follows that angle *F* is congruent to angle *J* and, therefore, the measure of angle *F* is equal to the measure of angle *J*. The sine ratios of angles of equal measure are equal. Since the measure of angle *F* is equal to the measure of angle *J*, $\sin F = \sin J$. It's given that $\sin F = \frac{308}{317}$. Therefore, $\sin J$ is $\frac{308}{317}$.

Choice A is incorrect. This is the value of $\cos J$, not the value of $\sin J$.

Choice C is incorrect. This is the reciprocal of the value of $\sin J$, not the value of $\sin J$.

Choice D is incorrect. This is the reciprocal of the value of $\cos J$, not the value of $\sin J$.

▼ 25 — B

Choice B is correct. Triangles ADB and CDB are both $30° - 60° - 90°$ triangles and share $\overline{BD}$. Therefore, triangles ADB and CDB are congruent by the angle-side-angle postulate. Using the properties of $30° - 60° - 90°$ triangles, the length of $\overline{AD}$ is half the length of hypotenuse $\overline{AB}$. Since the triangles are congruent, $AB = BC = 12$. So the length of $\overline{AD}$ is $\frac{12}{2} = 6$.

Alternate approach: Since angle CBD has a measure of $30°$, angle ABC must have a measure of $60°$. It follows that triangle ABC is equilateral, so side AC also has length 12. It also follows that the altitude BD is also a median, and therefore the length of AD is half of the length of AC, which is 6.

Choice A is incorrect. If the length of $\overline{AD}$ were 4, then the length of $\overline{AB}$ would be 8. However, this is incorrect because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12.

Choices C and D are also incorrect. Following the same procedures as used to test choice A gives $\overline{AB}$ a length of $12\sqrt{2}$ for choice C and $12\sqrt{3}$ for choice D. However, these results cannot be true because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12.

▼ 26 — A

Choice A is correct. For the right triangle shown, the lengths of the legs are a units and 6 units, and the length of the hypotenuse is 21 units. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + 6^2 = 21^2$. Subtracting 6^2 from both sides of this equation yields $a^2 = 21^2 - 6^2$. Taking the square root of both sides of this equation yields $a = \pm\sqrt{21^2 - 6^2}$. Since a is a length, a must be positive. Therefore, $a = \sqrt{21^2 - 6^2}$. Thus, for the triangle shown, $\sqrt{21^2 - 6^2}$ represents the value of a.

Choice B is incorrect. For the triangle shown, this expression represents the value of a^2, not a.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 27 — 565

The correct answer is 565. In triangle XYZ, angle Y is a right angle. Therefore, triangle XYZ is a right triangle. The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to that angle to the length of the hypotenuse. In right triangle XYZ, side YZ is the side adjacent to angle Z and side XZ is the hypotenuse. It's given that the length of side YZ is 560 and the length of side XZ is 565. Therefore, $\cos Z = \frac{560}{565}$. It's also given that $\cos Z = \frac{560}{c}$. It follows that $\frac{560}{c} = \frac{560}{565}$. Therefore, $c = 565$.

▼ 28 — D

Choice D is correct. The hypotenuse of triangle RST is the longest side and is across from the right angle. The longest side length given is 584, which is the length of side TR. Therefore, the hypotenuse of triangle RST is side TR, so the right angle is angle S. The

tangent of an acute angle in a right triangle is the ratio of the length of the opposite side, which is the side across from the angle, to the length of the adjacent side, which is the side closest to the angle that is not the hypotenuse. It follows that the opposite side of angle T is side RS and the adjacent side of angle T is side ST. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 440 for RS and 384 for ST in this equation yields $\tan T = \frac{440}{384}$. This is equivalent to $\tan T = \frac{55}{48}$. It's given that triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. It follows that R corresponds to U. Therefore, the hypotenuse of triangle UVW is side WU, which means $\tan W = \frac{UV}{VW}$. Since the lengths of corresponding sides of similar triangles are proportional, $\frac{RS}{ST} = \frac{UV}{VW}$. Therefore, $\tan W = \frac{UV}{VW}$ is equivalent to $\tan W = \frac{RS}{ST}$, or $\tan W = \tan T$. Thus, $\tan W = \frac{55}{48}$.

Choice A is incorrect. This is the value of $\cos W$, not $\tan W$.

Choice B is incorrect. This is the value of $\sin W$, not $\tan W$.

Choice C is incorrect. This is the value of $\frac{1}{\tan W}$, not $\tan W$.

▼ 29 — .3928, .3929, 11/28

The correct answer is $\frac{11}{28}$. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to the angle with measure $x°$ is 11 units and the length of the hypotenuse is 28 units. Therefore, the value of $\cos x°$ is $\frac{11}{28}$. Note that 11/28, .3928, .3929, 0.392, and 0.393 are examples of ways to enter a correct answer.

▼ 30 — .14, 7/50

The correct answer is $\frac{7}{50}$. It's given that triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. In similar triangles, the tangents of corresponding angles are equal. Since angle A and angle D are corresponding angles, if $\tan A = \frac{50}{7}$, then $\tan D = \frac{50}{7}$. It's also given that angles C and F are right angles. It follows that triangle DEF is a right triangle with acute angles D and E. The tangent of one acute angle in a right triangle is the inverse of the tangent of the other acute angle in the triangle. Therefore, $\tan E = \frac{1}{\tan D}$. Substituting $\frac{50}{7}$ for $\tan D$ in this equation yields $\tan E = \frac{1}{\frac{50}{7}}$, or $\tan E = \frac{7}{50}$. Thus, if $\tan A = \frac{50}{7}$, the value of $\tan E$ is $\frac{7}{50}$. Note that 7/50 and .14 are examples of ways to enter a correct answer.

▼ 31 — 38

The correct answer is 38. It's given that in rectangle *PQRS*, the length of side *PS* is 8 more than the length of side *RS*. In a rectangle, opposite sides are congruent. Since side *QR* is opposite side *PS*, it follows that side *QR* is congruent to side *PS*. Thus, the length of side *QR* is 8 more than the length of side *RS*. Let *x* represent the length of side *RS*. It follows that the length of side *QR* is represented by the expression $x + 8$. Since all angles in a rectangle are right angles, angle *R* is a right angle. It follows that side *QR*, side *RS*, and line segment *QS* form a right triangle, where line segment *QS* is the hypotenuse and sides *QR* and *RS* are the legs. It's given that the length of line segment *QS* is $\sqrt{2{,}344}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length *c* and legs with lengths *a* and *b*, then $a^2 + b^2 = c^2$. Substituting *x* for *a*, $x + 8$ for *b*, and $\sqrt{2{,}344}$ for *c* in this equation yields $x^2 + (x + 8)^2 = (\sqrt{2{,}344})^2$, or $x^2 + (x + 8)^2 = 2{,}344$. Expanding the left-hand side of this equation yields $x^2 + x^2 + 16x + 64 = 2{,}344$, or $2x^2 + 16x + 64 = 2{,}344$. Dividing both sides of this equation by 2 yields $x^2 + 8x + 32 = 1{,}172$. To complete the square, subtracting 16 from both sides of this equation yields $x^2 + 8x + 16 = 1{,}156$. Since the left-hand side of this equation is a perfect square trinomial, it follows that $x^2 + 8x + 16 = 1{,}156$ is equivalent to $(x + 4)^2 = 1{,}156$. Taking the positive square root of both sides of this equation yields $x + 4 = 34$. Subtracting 4 from both sides of this equation yields $x = 30$. Since the length of side *QR* is represented by the expression $x + 8$, it follows that the length of side *QR* is $30 + 8$, or 38.

▼ 32 — C

Choice C is correct. It's given that the logo is in the shape of a trapezoid that consists of three congruent equilateral triangles, and that the perimeter of the trapezoid is 20 centimeters (cm). Since the perimeter of the trapezoid is the sum of the lengths of 5 of the sides of the triangles, the length of each side of an equilateral triangle is $\frac{20}{5} = 4$ cm.

Dividing up one equilateral triangle into two right triangles yields a pair of congruent 30°-60°-90° triangles. The shorter leg of each right triangle is half the length of the side of an equilateral triangle, or 2 cm. Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, the height of the equilateral triangle can be found. Substituting $a = 2$ and $c = 4$ and solving for b yields $\sqrt{4^2 - 2^2} = \sqrt{12} = 2\sqrt{3}$ cm. The area of one equilateral triangle is $\frac{1}{2}bh$, where $b = 2$ and $h = 2\sqrt{3}$. Therefore, the area of one equilateral triangle is $\frac{1}{2}(4)(2\sqrt{3}) = 4\sqrt{3}$ cm^2. The shaded area consists of two such triangles, so its area is $(2)(4)\sqrt{3} = 8\sqrt{3}$ cm^2.

Alternate approach: The area of a trapezoid can be found by evaluating the expression $\frac{1}{2}(b_1+b_2)h$, where $b_1$is the length of one base, b_2 is the length of the other base, and h is the height of the trapezoid. Substituting $b_1=8$, $b_2=4$, and $h=2\sqrt{3}$ yields the expression $\frac{1}{2}(8+4)(2\sqrt{3})$, or $\frac{1}{2}(12)(2\sqrt{3})$, which gives an area of $12\sqrt{3}$ cm^2 for the trapezoid. Since two-thirds of the trapezoid is shaded, the area of the shaded region is $\frac{2}{3}\times 12\sqrt{3}=8\sqrt{3}$.

Choice A is incorrect. This is the height of the trapezoid. Choice B is incorrect. This is the area of one of the equilateral triangles, not two. Choice D is incorrect and may result from using a height of 4 for each triangle rather than the height of $2\sqrt{3}$.

▼ 33 — 36

The correct answer is 36. It's given that side *SQ* is the hypotenuse of right triangle *QRS*. It's also given that the length of side *SQ* is $4\sqrt{82}$ and the length of side *QR* is 4. It follows that sides *QR* and *RS* are the legs of triangle *QRS*. The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Let *b* represent the length of side *RS*. By the Pythagorean theorem, $4^2 + b^2 = (4\sqrt{82})^2$, or $16 + b^2 = 1{,}312$. Subtracting 16 from both sides of this equation yields $b^2 = 1{,}296$. Taking the positive square root of both sides of this equation yields $b = 36$. Therefore, the length of side *RS* is 36.

▼ 34 — D

Choice D is correct. Corresponding angles in similar triangles have equal measures. It's given that triangle *ABC* is similar to triangle *DEF*, where *A* corresponds to *D*, so the measure of angle *A* is equal to the measure of angle *D*. Therefore, if $\tan A = \sqrt{3}$, then $\tan D = \sqrt{3}$. It's given that angles *C* and *F* are right angles, so triangles *ABC* and *DEF* are right triangles. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle *D* is side *DF*. The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle *D* is side *EF*. The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan D = \frac{EF}{DF}$. If $DF = 125$, the length of side *EF* can be found by substituting $\sqrt{3}$ for $\tan D$ and 125 for *DF* in the equation $\tan D = \frac{EF}{DF}$, which yields $\sqrt{3} = \frac{EF}{125}$. Multiplying both

sides of this equation by 125 yields $125\sqrt{3} = EF$. Since the length of side EF is $\sqrt{3}$ times the length of side DF, it follows that triangle DEF is a special right triangle with angle measures $30°$, $60°$, and $90°$. Therefore, the length of the hypotenuse, $\overline{DE}$, is 2 times the length of side DF, or $DE = 2DF$. Substituting 125 for DF in this equation yields $DE = 2125$, or $DE = 250$. Thus, if $\tan A = \sqrt{3}$ and $DF = 125$, the length of $\overline{DE}$ is 250.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of $\overline{EF}$, not $\overline{DE}$.

▼ 35 — C

Choice C is correct. The area, A, of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is its corresponding height. It's given that the length of $\overline{XY}$ is 26 units. This length represents a base of the triangle, so $b = 26$. Let h represent its corresponding height. It's given that the measure of angle X is $54°$ and the length of $\overline{XZ}$ is 19 units. It follows that $\sin 54° = \frac{h}{19}$. Multiplying both sides of this equation by 19 yields $19\sin 54° = h$. Substituting 26 for b and $19\sin 54°$ for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(26)(19\sin 54°)$, or $A = 247\sin 54°$. Therefore, the area, in square units, of triangle XYZ is $247\sin 54°$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 36 — 12

The correct answer is 12. The diagonal of a rectangle forms a right triangle, where the shorter side and the longer side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. It's given that the length of the rectangle's diagonal is $3\sqrt{17}$ and the length of the rectangle's shorter side is 3. Thus, the length of the hypotenuse of the right triangle formed by the diagonal is $3\sqrt{17}$ and the length of one of the legs is 3. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting $3\sqrt{17}$ for c and 3 for b in this equation yields $a^2 + 3^2 = 3\sqrt{17}^2$, or $a^2 + 9 = 153$. Subtracting 9 from both sides of this equation yields $a^2 = 144$. Taking the square root of both sides of this equation yields

$a = \pm\sqrt{144}$, or $a = \pm 12$. Since a represents a length, which must be positive, the value of a is 12. Thus, the length of the rectangle's longer side is 12.

▼ 37 — A

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other angle, the angles are complementary. It follows that $\angle Q$ and $\angle R$ are complementary. That is, the sum of the measures of the angles is 90 degrees. It's given that the measure of $\angle Q$ is $x + 61$ degrees and the measure of $\angle R$ is $4x + 4$ degrees. It follows that $(x + 61) + (4x + 4) = 90$. By combining like terms, this equation can be rewritten as $5x + 65 = 90$. Subtracting 65 from each side of this equation yields $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\cos(Q) = \cos(R)$ rather than $\cos(Q) = \sin(R)$.

Choice C is incorrect. This would be the value of x if $\cos(Q) = -\cos(R)$ rather than $\cos(Q) = \sin(R)$ and if $\angle R$ were obtuse rather than acute.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 38 — C

Choice C is correct. The sine of an angle is equal to the cosine of its complementary angle. Since angles with measures $24°$ and $66°$ are complementary to each other, $\sin 24°$ is equal to $\cos 66°$ and $\sin 66°$ is equal to $\cos 24°$. Substituting $\cos 66°$ for $\sin 24°$ and $\cos 24°$ for $\sin 66°$ in the given expression yields $\cos 66° \cos 66° + \cos 24° \cos 24°$, or $\cos 66°^2 + \cos 24°^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 39 — ?

The correct answer is 12. It is given that the triangle has angle measures of 30°, 60°, and 90°, and so the triangle is a special right triangle. The side measures of this type of special triangle are in the ratio $2 : 1 : \sqrt{3}$. If x is the measure of the shortest leg, then the measure of

the other leg is $\sqrt{3}x$ and the measure of the hypotenuse is 2x. The perimeter of the triangle is given to be $18+6\sqrt{3}$, and so the equation for the perimeter can be written as $2x+x+\sqrt{3}x=18+6\sqrt{3}$. Combining like terms and factoring out a common factor of x on the left-hand side of the equation gives $(3+\sqrt{3})x=18+6\sqrt{3}$. Rewriting the right-hand side of the equation by factoring out 6 gives $(3+\sqrt{3})x=6(3+\sqrt{3})$. Dividing both sides of the equation by the common factor $(3+\sqrt{3})$ gives x = 6. The longest side of the right triangle, the hypotenuse, has a length of 2x, or 2(6), which is 12.

▼ 40 — A

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other, the angles are complementary. Therefore, the sum of the measures, in degrees, of $\angle Q$ and $\angle R$ is 90. It's given that the measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. Therefore, $(x + 52) + (4x + 13) = 90$. Combining like terms in this equation yields $5x + 65 = 90$. Subtracting 65 from each side of this equation results in $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\angle Q$ and $\angle R$ were congruent rather than complementary.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 41 — 66

The correct answer is 66. It's given that each vertex of the rectangle lies on the circumference of the circle. Therefore, the length of the diameter of the circle is equal to the length of the diagonal of the rectangle. The diagonal of a rectangle forms a right triangle with the shortest and longest sides of the rectangle, where the shortest side and the longest side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. Let s represent the length, in units, of the shortest side of the rectangle. Since it's given that the diagonal is twice the length of the shortest side, $2s$ represents the length, in units, of the diagonal of the rectangle. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting s for a and $2s$ for c in this equation yields $s^2 + b^2 = 2s^2$, or $s^2 + b^2 = 4s^2$. Subtracting s^2 from both sides of this equation yields $b^2 = 3s^2$. Taking the

positive square root of both sides of this equation yields $b = s\sqrt{3}$. Therefore, the length, in units, of the rectangle's longest side is $s\sqrt{3}$. The area of a rectangle is the product of the length of the shortest side and the length of the longest side. The lengths, in units, of the shortest and longest sides of the rectangle are represented by s and $s\sqrt{3}$, and it's given that the area of the rectangle is $1{,}089\sqrt{3}$ square units. It follows that $1{,}089\sqrt{3} = ss\sqrt{3}$, or $1{,}089\sqrt{3} = s^2\sqrt{3}$. Dividing both sides of this equation by $\sqrt{3}$ yields $1{,}089 = s^2$. Taking the positive square root of both sides of this equation yields $33 = s$. Since the length, in units, of the diagonal is represented by $2s$, it follows that the length, in units, of the diagonal is 233, or 66. Since the length of the diameter of the circle is equal to the length of the diagonal of the rectangle, the length, in units, of the diameter of the circle is 66.

▼ 42 — C

Choice C is correct. In the triangle shown, the measure of angle B is $30°$ and angle C is a right angle, which means that it has a measure of $90°$. Since the sum of the angles in a triangle is equal to $180°$, the measure of angle A is equal to $180° - 30 + 90°$, or $60°$. In a right triangle whose acute angles have measures $30°$ and $60°$, the lengths of the legs can be represented by the expressions x, $x\sqrt{3}$, and $2x$, where x is the length of the leg opposite the angle with measure $30°$, $x\sqrt{3}$ is the length of the leg opposite the angle with measure $60°$, and $2x$ is the length of the hypotenuse. In the triangle shown, the hypotenuse has a length of 54. It follows that $2x = 54$, or $x = 27$. Therefore, the length of the leg opposite angle B is 27 and the length of the leg opposite angle A is $27\sqrt{3}$. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. The length of the leg opposite angle A is $27\sqrt{3}$ and the length of the leg adjacent to angle A is 27. Therefore, the value of $\tan A$ is $\frac{27\sqrt{3}}{27}$, or $\sqrt{3}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\frac{1}{\tan A}$, not the value of $\tan A$.

Choice D is incorrect. This is the length of the leg opposite angle A, not the value of $\tan A$.

▼ 43 — D

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure

that the length of side RS is 25. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS, an expression representing QS can be found by substituting 25 for RS in this equation, which yields $\sin Q = \frac{25}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 25$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{25}{\sin Q}$. Therefore, the expression $\frac{25}{\sin Q}$ represents the length of $\overline{QS}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 44 — 284/3, 94.66, 94.67

The correct answer is $\frac{284}{3}$. Since the perimeter of a triangle is the sum of the lengths of its sides, and the given triangle is equilateral, the length of each side is $\frac{852}{3}$, or 284, centimeters (cm). Right triangle AMO can be formed, where M is the midpoint of one of the triangle's sides, A is one of this side's endpoints, and O is the center of the circle. It follows that AM is $\frac{284}{2}$, or 142, cm. Additionally, triangle AMO has angles measuring $30°$, $60°$, and $90°$, where the measure of angle OMA is $90°$ and the measure of angle OAM is $30°$. It follows that the length of side MO is half the length of hypotenuse AO, and the length of side AM is $\sqrt{3}$ times the length of side MO. It's given that $AO = w\sqrt{3}$ cm. Therefore, $MO = \frac{w\sqrt{3}}{2}$ cm and $AM = \frac{w\sqrt{3}\sqrt{3}}{2}$ cm, which is equivalent to $AM = \frac{3w}{2}$ cm. Since $AM = 142$ cm, it follows that $\frac{3w}{2} = 142$. Multiplying both sides of this equation by 2 yields $3w = 284$. Dividing both sides of this equation by 3 yields $w = \frac{284}{3}$. Note that 284/3, 94.66, and 94.67 are examples of ways to enter a correct answer.

▼ 45 — ?

The correct answer is 6. Since $\tan B = \frac{3}{4}$, $\triangle ABC$ and $\triangle DBE$ are both similar to 3-4-5 triangles. This means that they are both similar to the right triangle with sides of lengths 3, 4, and 5. Since $BC = 15$, which is 3 times as long as the hypotenuse of the 3-4-5 triangle, the similarity ratio of $\triangle ABC$ to the 3-4-5 triangle is 3:1. Therefore, the length of $\overline{AC}$ (the side opposite to $\angle B$) is $3 \times 3 = 9$, and the length of $\overline{AB}$ (the side adjacent to $\angle B$) is $4 \times 3 = 12$. It is also given that $DA = 4$. Since $AB = DA + DB$ and $AB = 12$, it follows that $DB = 8$, which means that the similarity ratio of $\triangle DBE$ to the 3-4-5 triangle is 2:1 ($\overline{DB}$ is

the side adjacent to $\angle B$). Therefore, the length of $\overline{DE}$, which is the side opposite to $\angle B$, is $3 \times 2 = 6$.

▼ 46 — C

Choice C is correct. The perimeter of a triangle is the sum of the lengths of its sides. Since the given triangle is an isosceles right triangle, the length of each leg is the same and the length of the hypotenuse is equal to $\sqrt{2}$ times the length of a leg. Let x represent the length, in inches, of a leg of this isosceles right triangle. Therefore, the perimeter, in inches, of the triangle is $x + x + x\sqrt{2}$, or $2x + x\sqrt{2}$, which is equivalent to $x(2 + \sqrt{2})$. It's given that the perimeter of this triangle is $18 + 18\sqrt{2}$ inches. Thus, $x2 + \sqrt{2} = 18 + 18\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $x = \frac{18 + 18\sqrt{2}}{2 + \sqrt{2}}$. Multiplying the right-hand side of this equation by $\frac{2 - \sqrt{2}}{2 - \sqrt{2}}$ yields $x = \frac{36 + 36\sqrt{2} - 18\sqrt{2} - 36}{2}$, or $x = 9\sqrt{2}$. It follows that the length, in inches, of a leg of this isosceles right triangle is $9\sqrt{2}$. Therefore, the length, in inches, of the hypotenuse of this isosceles right triangle is $9\sqrt{2}\sqrt{2}$, or 18.

Choice A is incorrect. If this were the length of the hypotenuse, the perimeter would be $9 + 9\sqrt{2}$ inches.

Choice B is incorrect. This is the length, in inches, of a leg of this triangle, not the hypotenuse.

Choice D is incorrect. If this were the length of the hypotenuse, the perimeter would be $36 + 18\sqrt{2}$ inches.

▼ 47 — 338

The correct answer is 338. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. In triangle XYZ, it's given that angle Y is a right angle. Thus, $\overline{XY}$ is the leg opposite of angle Z and $\overline{YZ}$ is the leg adjacent to angle Z. It follows that $\tan Z = \frac{XY}{YZ}$. It's also given that the measure of angle Z is $33°$ and the length of $\overline{YZ}$ is 26 units. Substituting $33°$ for Z and 26 for YZ in the equation $\tan Z = \frac{XY}{YZ}$ yields $\tan 33° = \frac{XY}{26}$. Multiplying each side of this equation by 26 yields $26\tan 33° = XY$. Therefore, the length of $\overline{XY}$ is $26\tan 33°$. The area of a triangle is half the product of the lengths of its legs. Since the length of $\overline{YZ}$ is 26 and the length of $\overline{XY}$ is $26\tan 33°$, it follows that the area of triangle XYZ is $\frac{1}{2}2626\tan 33°$ square units, or $338\tan 33°$ square units. It's given that the area, in square units, of triangle XYZ can be represented by the expression $k\tan 33°$, where k is a constant. Therefore, 338 is the value of k.

▼ 48 — D

Choice D is correct. The tangent of a nonright angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Using that definition for tangent, in a right triangle with legs that have lengths a and b, the tangent of one acute angle is $\frac{a}{b}$ and the tangent for the other acute angle is $\frac{b}{a}$. It follows that the tangents of the acute angles in a right triangle are reciprocals of each other. Therefore, the tangent of the other acute angle in the given triangle is the reciprocal of $\frac{\sqrt{3}}{3}$ or $\frac{3}{\sqrt{3}}$.

Choice A is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the tangent of the angle described. Choice B is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the reciprocal of the tangent of the angle described. Choice C is incorrect and may result from interpreting the tangent of the other acute angle as equal to the tangent of the angle described.

▼ 49 — D

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS, an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of QS.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 50 — 2.4, 12/5

The correct answer is $\frac{12}{5}$. It's given that in right triangle XYZ, $\angle X$ and $\angle Z$ are acute angles. It follows that $\angle Y$ is a right angle. It's given that point R lies on XZ such that YR is perpendicular to XZ. Since $\angle Y$ is a right angle and is comprised of $\angle XYR$ and $\angle ZYR$, it follows that the sum of the measures of $\angle XYR$ and $\angle ZYR$ is $90°$. It's given that the measure of $\angle XYR$ is $b°$. It follows that the measure of $\angle ZYR$ is $(90 - b)°$. Since YR is perpendicular to XZ, it follows that $\angle XRY$ and $\angle ZRY$ are right angles. It follows that ΔXRY and ΔZRY are right triangles. It's given that the measure of $\angle XZY$ is $a°$. Because the sum of the measures of the interior angles of a triangle is $180°$, it follows that the measures of $\angle ZRY$, $\angle XZY$, and $\angle ZYR$ have a sum of $180°$. Since the measure of $\angle ZRY$ is $90°$, the measure of $\angle YZR$ is $a°$, and the measure of $\angle ZYR$ is $(90 - b)°$, it follows that $90 + a + 90 - b = 180$, or $a - b + 180 = 180$. Subtracting 180 from both sides of this equation yields $a - b = 0$. Adding b to both sides of this equation yields $a = b$. It's given that $\sin a° = \frac{12}{13}$. Since $a = b$, it follows that $\sin b° = \frac{12}{13}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. It follows that, for a positive constant k, $12k$ represents the length of the leg opposite $\angle XYR$ and $13k$ represents the length of the hypotenuse of $\triangle XRY$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. Let nk represent the length of the leg adjacent to $\angle XYR$. It follows that $(13k)^2 = (12k)^2 + (nk)^2$, or $169k^2 = 144k^2 + n^2k^2$. Subtracting $144k^2$ from both sides of this equation yields $25k^2 = n^2k^2$. Taking the positive square root of both sides of this equation yields $5k = nk$. Therefore, the length of the leg adjacent to $\angle XYR$ is $5k$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan b° = \frac{12k}{5k}$, or $\tan b° = \frac{12}{5}$. Note that 12/5 and 2.4 are examples of ways to enter a correct answer.

▼ 51 — B

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that in triangle ABC, angle B is a right angle. Therefore, triangle ABC is a right triangle, where the hypotenuse is side AC and the legs are sides AB and BC. It's given that the lengths of sides AB and BC are $10\sqrt{37}$ and $24\sqrt{37}$, respectively. Substituting these values for a and b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$, which is equivalent to $c^2 = 10037 + 57637$, or $c^2 = 67637$. Taking the square root of both sides of this

equation yields $c = \pm 26\sqrt{37}$. Since c represents the length of the hypotenuse, side AC, c must be positive. Therefore, the length of side AC is $26\sqrt{37}$.

Choice A is incorrect. This is the result of solving the equation $c = 24\sqrt{37} - 10\sqrt{37}$, not $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$.

Choice C is incorrect. This is the result of solving the equation $c = 10\sqrt{37} + 24\sqrt{37}$, not $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 52 — B

Choice B is correct. It's given that the right triangle is isosceles. In an isosceles right triangle, the two legs have equal lengths, and the length of the hypotenuse is $\sqrt{2}$ times the length of one of the legs. Let l represent the length, in inches, of each leg of the isosceles right triangle. It follows that the length of the hypotenuse is $l\sqrt{2}$ inches. The perimeter of a figure is the sum of the lengths of the sides of the figure. Therefore, the perimeter of the isosceles right triangle is $l + l + l\sqrt{2}$ inches. It's given that the perimeter of the triangle is $94 + 94\sqrt{2}$ inches. It follows that $l + l + l\sqrt{2} = 94 + 94\sqrt{2}$. Factoring the left-hand side of this equation yields $1 + 1 + \sqrt{2}l = 94 + 94\sqrt{2}$, or $2 + \sqrt{2}l = 94 + 94\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $l = \frac{94 + 94\sqrt{2}}{2 + \sqrt{2}}$. Rationalizing the denominator of the right-hand side of this equation by multiplying the right-hand side of the equation by $\frac{2 - \sqrt{2}}{2 - \sqrt{2}}$ yields $l = \frac{94 + 94\sqrt{2}2 - \sqrt{2}}{2 + \sqrt{2}2 - \sqrt{2}}$. Applying the distributive property to the numerator and to the denominator of the right-hand side of this equation yields $l = \frac{188 - 94\sqrt{2} + 188\sqrt{2} - 94\sqrt{4}}{4 - 2\sqrt{2} + 2\sqrt{2} - \sqrt{4}}$. This is equivalent to $l = \frac{94\sqrt{2}}{2}$, or $l = 47\sqrt{2}$. Therefore, the length, in inches, of one leg of the isosceles right triangle is $47\sqrt{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length, in inches, of the hypotenuse.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 53 — D

Choice D is correct. It's given that in triangle XYZ, the measure of angle X is $90°$, so triangle XYZ is a right triangle. It's also given that point W lies on segment YZ, and segment WX is perpendicular to segment YZ. It follows that triangle WYX is a right triangle. Triangle WYX and triangle XYZ are right triangles with the same interior angle measure at point Y. It follows that triangle WYX and triangle XYZ are similar, where angles W, Y, and

X in triangle WYX correspond to angles X, Y, and Z, respectively, in triangle XYZ. Since corresponding angles in similar triangles are congruent, their tangents have equal value. Thus, the tangent of angle Z is equal to the tangent of angle X in triangle WYX. The opposite side of angle X in triangle WYX is segment WY, and the adjacent side is segment WX. It's given that the length of segment WY is 572 and the length of segment WX is 429. It follows that the tangent of angle X in triangle WYX is $\frac{572}{429}$, or $\frac{4}{3}$. Therefore, the value of $\tan Z$ is $\frac{4}{3}$.

Choice A is incorrect. This is the value of $\cos Z$, not $\tan Z$.

Choice B is incorrect. This is the value of $\tan Y$, not $\tan Z$.

Choice C is incorrect. This is the value of $\sin Z$, not $\tan Z$.

▼ 54 — .6956, .6957, 16/23

The correct answer is $\frac{16}{23}$. In a right triangle, the sine of an acute angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the length of the side opposite the angle with measure $x°$ is 16 units and the length of the hypotenuse is 23 units. Therefore, the value of $\sin x°$ is $\frac{16}{23}$. Note that 16/23, .6956, .6957, 0.695, and 0.696 are examples of ways to enter a correct answer.

▼ 55 — ?

The correct answer is 0. Note that no matter where point W is on $\overline{RT}$, the sum of the measures of $\angle RSW$ and $\angle WST$ is equal to the measure of $\angle RST$, which is $90°$. Thus, $\angle RSW$ and $\angle WST$ are complementary angles. Since the cosine of an angle is equal to the sine of its complementary angle, $\cos(\angle RSW) = \sin(\angle WST)$. Therefore, $\cos(\angle RSW) - \sin(\angle WST) = 0$.

▼ 56 — .8823, .8824, 15/17

The correct answer is $\frac{15}{17}$. It's given that angle J is the right angle in triangle JKL. Therefore, the acute angles of triangle JKL are angle K and angle L. The hypotenuse of a right triangle is the side opposite its right angle. Therefore, the hypotenuse of triangle JKL is side KL. The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to the angle to the length of the hypotenuse. It's given that $\cos K = \frac{24}{51}$. This can be written as $\cos K = \frac{8}{17}$. Since the cosine of angle K is a ratio, it follows that the length of the side adjacent to angle K is $8n$ and the length of the hypotenuse is $17n$, where n is a constant.

Therefore, $JK = 8n$ and $KL = 17n$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. For triangle JKL, it follows that $JK^2 + JL^2 = KL^2$. Substituting $8n$ for JK and $17n$ for KL yields $8n^2 + JL^2 = 17n^2$. This is equivalent to $64n^2 + JL^2 = 289n^2$. Subtracting $64n^2$ from each side of this equation yields $JL^2 = 225n^2$. Taking the square root of each side of this equation yields $JL = 15n$. Since $\cos L = \frac{JL}{KL}$, it follows that $\cos L = \frac{15n}{17n}$, which can be rewritten as $\cos L = \frac{15}{17}$. Note that 15/17, .8824, .8823, and 0.882 are examples of ways to enter a correct answer.

▼ 57 — 104

The correct answer is 104. An equilateral triangle is a triangle in which all three sides have the same length and all three angles have a measure of $60°$. The height of the triangle, $k\sqrt{3}$, is the length of the altitude from one vertex. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the $60°$ angle in each 30-60-90 right triangle. Since the altitude has a length of $k\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each $30°$ angle has a length of k and each hypotenuse has a length of $2k$. In this case, the hypotenuse of each 30-60-90 right triangle is a side of the equilateral triangle; therefore, each side length of the equilateral triangle is $2k$. The perimeter of a triangle is the sum of the lengths of each side. It's given that the perimeter of the equilateral triangle is 624; therefore, $2k + 2k + 2k = 624$, or $6k = 624$. Dividing both sides of this equation by 6 yields $k = 104$.

▼ 58 — B

Choice B is correct. It's given that angle Z in triangle XYZ is a right angle. Thus, side YZ is the leg opposite angle X and side XZ is the leg adjacent to angle X. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan X = \frac{YZ}{XZ}$. It's given that $\tan X = \frac{12}{35}$ and the length of side YZ is 24 units. Substituting $\frac{12}{35}$ for $\tan X$ and 24 for YZ in the equation $\tan X = \frac{YZ}{XZ}$ yields $\frac{12}{35} = \frac{24}{XZ}$. Multiplying both sides of this equation by $35XZ$ yields $12XZ = 2435$, or $12XZ = 840$. Dividing both sides of this equation by 12 yields $XZ = 70$. The length XY can be calculated using the Pythagorean theorem, which states that if a right triangle has legs with lengths of a and b and a hypotenuse with length c, then $a^2 + b^2 = c^2$. Substituting 70 for a and 24 for b in this equation yields $70^2 + 24^2 = c^2$, or $5{,}476 = c^2$. Taking the square root of both sides of this equation yields $\pm 74 = c$. Since the length of the hypotenuse must be positive, $74 = c$. Therefore, the length of XY is 74 units. The perimeter of a triangle is the

sum of the lengths of all sides. Thus, $74 + 70 + 24$ units, or 168 units, is the perimeter of triangle XYZ.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the perimeter, in units, for a right triangle where the length of side YZ is 12 units, not 24 units.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 59 — .9811, 52/53

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is 53. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let x represent TS, the length of the other leg of the triangle shown. Therefore, $53^2 = (\sqrt{105})^2 + x^2$, or $2{,}809 = 105 + x^2$. Subtracting 105 from both sides of this equation yields $2{,}704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, TS, the length of the other leg of the triangle shown, is 52. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle R is 52, and the length of the hypotenuse is 53. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that 52/53 or .9811 are examples of ways to enter a correct answer.

▼ 60 — A

Choice A is correct. When a square is inscribed in a circle, a diagonal of the square is a diameter of the circle. It's given that a square is inscribed in a circle and the length of a radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. Therefore, the length of a diameter of the circle is $2\frac{20\sqrt{2}}{2}$ inches, or $20\sqrt{2}$ inches. It follows that the length of a diagonal of the square is $20\sqrt{2}$ inches. A diagonal of a square separates the square into two right triangles in which the legs are the sides of the square and the hypotenuse is a diagonal. Since a square has 4 congruent sides, each of these two right triangles has congruent legs and a hypotenuse of length $20\sqrt{2}$ inches. Since each of these two right triangles has congruent legs, they are both 45-45-90 triangles. In a 45-45-90 triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg. Let s represent the length of a leg of one of these 45-45-90 triangles. It follows that $20\sqrt{2} = \sqrt{2}s$. Dividing both sides of this equation by $\sqrt{2}$ yields $20 = s$. Therefore, the length of a leg of one of these 45-45-90 triangles is 20 inches. Since the legs of these two 45-45-90

triangles are the sides of the square, it follows that the side length of the square is 20 inches.

Choice B is incorrect. This is the length of a radius, in inches, of the circle.

Choice C is incorrect. This is the length of a diameter, in inches, of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 61 — ?

The correct answer is 24. The sine of an acute angle in a right triangle is equal to the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the sine of angle B, or $\sin(B)$, is equal to the ratio of the length of side $\overline{AC}$ to the length of side $\overline{AB}$. It's given that the length of side $\overline{AB}$ is 26 and that $\sin(B) = \frac{5}{13}$. Therefore, $\frac{5}{13} = \frac{AB}{26}$. Multiplying both sides of this equation by 26 yields $AC = 10$.

By the Pythagorean Theorem, the relationship between the lengths of the sides of triangle ABC is as follows: $26^2 = 10^2 + BC^2$, or $676 = 100 + BC^2$. Subtracting 100 from both sides of $676 = 100 + BC^2$ yields $576 = BC^2$. Taking the square root of both sides of $576 = BC^2$ yields $24 = BC$.

▼ 62 — 6.5, 13/2

The correct answer is 6.5. The figure shows that $\triangle BAC$ is a right triangle. In the figure, $\overline{AD}$ is perpendicular to $\overline{BC}$, which forms two additional right triangles: $\triangle BDA$ and $\triangle ADC$. If two pairs of corresponding angles in two triangles are congruent, then the triangles are similar. $\triangle BDA$ shares angle B with $\triangle BAC$ and $\triangle ADC$ shares angle C with $\triangle BAC$. Therefore, the figure contains three similar right triangles: $\triangle BAC$, $\triangle BDA$, and $\triangle ADC$. In similar triangles, the tangents of corresponding angles are equivalent. It's given that the tangent of angle CAD is 1.3. Since angle CAD in $\triangle ADC$ corresponds to angle ABD in $\triangle BDA$, it follows that the tangent of angle ABD is also 1.3. The tangent of an acute angle in a right triangle is equal to the length of the leg opposite the angle divided by the length of the leg adjacent to the angle. Therefore, $\tan ABD = \frac{AD}{BD}$. It's given that the length of $\overline{BD}$ is 5. Substituting 5 for BD and 1.3 for $\tan ABD$ in the equation $\tan ABD = \frac{AD}{BD}$ yields $1.3 = \frac{AD}{5}$. Multiplying both sides of this equation by 5 yields $6.5 = AD$. Therefore, the length of $\overline{AD}$ is 6.5. Note that 6.5 and 13/2 are examples of ways to enter a correct answer.

▼ 63 — .7241, 21/29

The correct answer is $\frac{21}{29}$. It's given that triangle *ABC* is similar to triangle *DEF*, where angle *A* corresponds to angle *D* and angles *C* and *F* are right angles. In similar triangles, the tangents of corresponding angles are equal. Therefore, if $\tan A = \frac{21}{20}$, then $\tan D = \frac{21}{20}$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Therefore, in triangle *DEF*, if $\tan D = \frac{21}{20}$, the ratio of the length of $\bar{EF}$ to the length of $\bar{DF}$ is $\frac{21}{20}$. If the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then the ratio of the length of $\bar{EF}$ to the length of $\bar{DF}$ is $\frac{21}{20}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. Therefore, the value of $\sin D$ is the ratio of the length of $\bar{EF}$ to the length of $\bar{DE}$. The length of $\bar{DE}$ can be calculated using the Pythagorean theorem, which states that if the lengths of the legs of a right triangle are *a* and *b* and the length of the hypotenuse is *c*, then $a^2 + b^2 = c^2$. Therefore, if the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then $21^2 + 20^2 = DE^2$, or $841 = DE^2$. Taking the positive square root of both sides of this equation yields $29 = DE$. Therefore, if the lengths of $\bar{EF}$ and $\bar{DF}$ are 21 and 20, respectively, then the length of $\bar{DE}$ is 29 and the ratio of the length of $\bar{EF}$ to the length of $\bar{DE}$ is $\frac{21}{29}$. Thus, if $\tan A = \frac{21}{20}$, the value of $\sin D$ is $\frac{21}{29}$. Note that 21/29, .7241, and 0.724 are examples of ways to enter a correct answer.

▼ 64 — B

Choice B is correct. It's given that right triangle *RST* is similar to triangle *UVW*, where *S* corresponds to *V* and *T* corresponds to *W*. It's given that the side lengths of the right triangle *RST* are $RS = 20$, $ST = 48$, and $TR = 52$. Corresponding angles in similar triangles are equal. It follows that the measure of angle *T* is equal to the measure of angle *W*. The hypotenuse of a right triangle is the longest side. It follows that the hypotenuse of triangle *RST* is side *TR*. The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle *S* is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle *T* is side *ST*. The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle *T* is side *RS*. The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 20 for *RS* and 48 for *ST* in this equation yields $\tan T = \frac{20}{48}$, or $\tan T = \frac{5}{12}$. The tangents of two acute angles with equal measures are equal. Since the measure of angle *T* is equal to the measure of angle *W*, it follows that $\tan T = \tan W$. Substituting $\frac{5}{12}$ for $\tan T$ in this equation yields $\frac{5}{12} = \tan W$. Therefore, the value of $\tan W$ is $\frac{5}{12}$.

Choice A is incorrect. This is the value of $\sin W$.

Choice C is incorrect. This is the value of $\cos W$.

Choice D is incorrect. This is the value of $\frac{1}{\tan W}$.

▼ 65 — ?

The correct answer is 30. In the figure given, since $\overline{BD}$ is parallel to $\overline{AE}$ and both segments are intersected by $\overline{CE}$, then angle BDC and angle AEC are corresponding angles and therefore congruent. Angle BCD and angle ACE are also congruent because they are the same angle. Triangle BCD and triangle ACE are similar because if two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. Since triangle BCD and triangle ACE are similar, their corresponding sides are proportional. So in triangle BCD and triangle ACE, $\overline{BD}$ corresponds to $\overline{AE}$ and $\overline{CD}$ corresponds to $\overline{CE}$. Therefore, $\frac{BD}{CD} = \frac{AE}{CE}$. Since triangle BCD is a right triangle, the Pythagorean theorem can be used to give the value of CD: $6^2 + 8^2 = CD^2$. Taking the square root of each side gives $CD = 10$. Substituting the values in the proportion $\frac{BD}{CD} = \frac{AE}{CE}$ yields $\frac{6}{10} = \frac{18}{CE}$. Multiplying each side by CE, and then multiplying by $\frac{10}{6}$ yields $CE = 30$. Therefore, the length of $\overline{CE}$ is 30.

▼ 66 — D

Choice D is correct. It's given that acute angle P has a measure of p radians and that acute angle T has a measure of t radians. If $\sin p = \cos t$, it follows that $p + t = \frac{\pi}{2}$, or $p = \frac{\pi}{2} - t$. It's also given that function g is defined by $g(x) = p(8)^x + 6t$. To determine the value of $g(0)$ in terms of t, substituting 0 for x and $\frac{\pi}{2} - t$ for p in the equation $g(x) = p(8)^x + 6t$ yields $g(0) = \frac{\pi}{2} - t(8)^0 + 6t$, or $g(0) = \frac{\pi}{2} - t(1) + 6t$. This is equivalent to $g(0) = \frac{\pi}{2} - t + 6t$, or $g(0) = 5t + \frac{\pi}{2}$. Therefore, $5t + \frac{\pi}{2}$ represents the value of $g(0)$ in terms of t.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 67 — 113

The correct answer is 113. It's given that the legs of a right triangle have lengths 24 centimeters and 21 centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if h represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1{,}017$. Taking the square root of each side of this equation yields $h = \sqrt{1{,}017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of d is 113.

▼ 68 — C

Choice C is correct. It's given that the equilateral triangle has a height of $39\sqrt{3}$. In equilateral triangles, all three angles have a measure of $60°$ and the height is equal to the length of an altitude from one vertex to the midpoint of the side across from the vertex. Thus, an altitude in this equilateral triangle has a length of $39\sqrt{3}$. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the $60°$ angle in each 30-60-90 right triangle. Since the altitude has a length of $39\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each $30°$ angle has a length of 39 and each hypotenuse has a length of $2(39)$, or 78. Therefore, each side of the equilateral triangle has a length of 78. In equilateral triangles, all three sides have equal length. Therefore, the perimeter of this equilateral triangle is $78 + 78 + 78$, or 234.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 69 — C

Choice C is correct. Since the triangle is an isosceles right triangle, the two sides that form the right angle must be the same length. Let x be the length, in inches, of each of those sides. The Pythagorean theorem states that in a right triangle, $a^2 + b^2 = c^2$, where c is the length of the hypotenuse and a and b are the lengths of the other two sides. Substituting x for a, x for b, and 58 for c in this equation yields $x^2 + x^2 = 58^2$, or $2x^2 = 58^2$. Dividing each side of this equation by 2 yields $x^2 = \frac{58^2}{2}$, or $x^2 = \frac{2 \cdot 58^2}{4}$. Taking the square root of each side of this equation yields two solutions: $x = \frac{58\sqrt{2}}{2}$ and $x = -\frac{58\sqrt{2}}{2}$. The value of x must be positive because it represents a side length. Therefore, $x = \frac{58\sqrt{2}}{2}$, or $x = 29\sqrt{2}$. The perimeter, in

inches, of the triangle is $58 + x + x$, or $58 + 2x$. Substituting $29\sqrt{2}$ for x in this expression gives a perimeter, in inches, of $58 + 229\sqrt{2}$, or $58 + 58\sqrt{2}$.

Choice A is incorrect. This is the length, in inches, of each of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice B is incorrect. This is the sum of the lengths, in inches, of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.